

ORLANDO SANFORD, FL, USA

WMO: 722057

Lat: 28.780N

Lon: 81.244W

Elev: 55

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12854

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 37.2	(c) 41.3	(d) 19.3	(e) 14.6	(f) 46.0	(g) 25.2	(h) 19.3	(i) 48.6	(j) 23.8	(k) 60.9	(l) 20.8	(m) 62.9	(n) 7.6	(o) 300	(p) 0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	94.4	75.9	92.9	75.8	91.3	75.7	79.2	88.0	78.4	87.0	78.0	86.5	8.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	141.6	81.9	76.4	137.9	81.6	75.6	134.3	81.2	42.6	88.1	41.9	87.2	41.3	86.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.0	(b) 17.9	(c) 16.1	(e) 31.2	(f) 97.2	(g) 3.5	(h) 1.8	(i) 28.7	(j) 98.5	(k) 26.7	(l) 99.5	(m) 24.8	(n) 100.5	(o) 22.3	(p) 101.8		
(4) 20.0	(5) 17.9	(6) 16.1	(7) 31.2	(8) 97.2	(9) 3.5	(10) 1.8	(11) 28.7	(12) 98.5	(13) 26.7	(14) 99.5	(15) 24.8	(16) 100.5	(17) 22.3	(18) 101.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.9	59.8	63.1	66.7	72.1	77.5	81.5	83.0	83.0	80.9	75.3	67.8	63.3
	DBStd	9.91	8.37	7.91	6.89	5.18	3.96	2.90	2.34	2.22	2.44	5.30	6.49	8.13
	HDD50	35	19	7	1	0	0	0	0	0	0	0	0	8
	HDD65	578	201	117	66	11	0	0	0	0	0	6	46	131
	CDD50	8389	322	374	519	662	854	945	1024	1023	927	785	535	419
	CDD65	3458	40	64	119	223	389	495	559	558	477	326	131	77
	CDH74	29856	226	408	851	1725	3417	4526	5592	5542	4155	2322	704	388
	CDH80	11153	17	62	183	513	1395	1915	2393	2280	1548	684	118	45
(14) Wind	WSAvg	7.1	7.6	7.7	8.3	8.0	7.6	6.6	6.0	6.1	6.4	7.0	7.1	7.0
(15) Precipitation	PrecAvg	51.7	2.0	2.0	3.1	2.1	3.6	8.0	7.2	7.6	7.3	3.9	2.0	2.1
	PrecMax	65.5	5.7	3.7	8.9	4.6	18.3	17.1	11.8	18.2	15.3	11.1	7.1	9.3
	PrecMin	38.7	0.0	0.4	0.0	1.1	0.9	5.0	2.1	2.6	3.9	0.7	0.1	0.4
	PrecStd	8.3	1.5	1.2	2.8	0.9	3.9	3.1	2.6	4.1	3.7	2.7	1.6	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.1	84.8	88.0	90.4	94.8	97.0	96.6	95.5	93.2	90.5	86.3	83.5
		MCWB	68.7	69.6	70.2	70.8	71.9	75.6	76.7	76.8	76.3	75.9	72.9	71.5
	2%	DB	79.4	81.9	84.3	87.9	91.9	93.5	93.9	93.6	91.1	88.3	83.2	81.3
		MCWB	66.6	68.1	68.3	70.2	71.8	75.7	76.5	76.9	75.8	75.1	71.4	69.7
	5%	DB	76.6	79.4	81.9	85.4	90.0	91.5	92.4	92.0	89.9	86.2	81.0	78.8
		MCWB	65.5	66.5	67.2	69.0	71.6	75.3	76.6	77.0	75.8	73.6	69.5	68.5
	10%	DB	73.2	75.9	79.3	82.8	87.8	89.8	90.6	90.4	88.1	83.9	78.6	75.3
		MCWB	64.4	64.9	65.6	67.9	71.4	75.3	76.7	77.0	75.6	72.4	67.8	67.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.2	72.4	73.4	74.9	76.5	79.3	80.1	80.3	79.6	78.3	75.5	73.5
		MCDB	78.2	79.1	81.1	83.5	85.9	88.7	89.3	89.1	87.2	85.4	81.4	79.8
	2%	WB	69.3	70.4	71.3	73.2	75.3	78.1	78.9	79.2	78.4	77.2	73.8	71.9
		MCDB	75.8	77.5	79.1	81.8	84.5	87.6	88.4	87.9	85.9	84.1	79.3	77.7
	5%	WB	67.6	68.8	69.8	71.9	74.4	77.3	78.2	78.4	77.8	76.1	72.2	70.4
		MCDB	73.9	76.1	77.9	80.7	83.5	86.7	87.3	86.8	85.3	82.6	77.7	75.7
	10%	WB	65.8	67.1	68.2	70.5	73.4	76.6	77.6	78.0	77.1	75.1	70.3	68.7
		MCDB	71.8	74.1	76.2	79.1	82.3	85.4	86.4	86.2	84.3	81.1	76.0	74.0
(35) Mean Daily Temperature Range	MDBR		20.2	20.3	20.8	20.5	19.7	16.8	16.5	15.8	15.1	16.5	18.4	18.8
	5% DB	MCDBR	21.1	21.7	22.2	21.7	21.9	19.1	18.1	17.5	16.8	16.8	18.6	19.1
		MCWBR	10.7	9.7	8.9	7.7	6.4	5.8	5.5	5.3	5.4	6.0	8.6	9.2
	5% WB	MCDBR	18.8	19.1	19.3	19.6	18.6	16.7	16.5	15.9	14.8	14.5	16.0	17.0
		MCWBR	11.0	9.9	9.3	8.1	6.6	6.0	5.7	5.5	5.4	5.7	8.6	9.7
(40) Clear-Sky Solar Irradiance	taub		0.339	0.353	0.376	0.394	0.427	0.457	0.480	0.470	0.423	0.397	0.358	0.347
	taud		2.525	2.496	2.424	2.353	2.306	2.293	2.226	2.269	2.396	2.436	2.497	2.524
	Ebn at Noon		286	291	290	286	275	266	259	261	271	272	277	278
	Edh at Noon		28	32	36	40	42	42	45	43	37	33	29	27
(44) All-Sky Solar Radiation	RadAvg		1064	1268	1618	1931	2008	1826	1822	1729	1537	1385	1132	971
	RadStd		79	91	124	101	135	167	106	115	97	99	62	77

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.76	N/A	N/A	N/A	+0.40	+1.01	-13	-131	+337
(47) Regional (43 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	-11	-85	+270

Nomenclature: See separate page